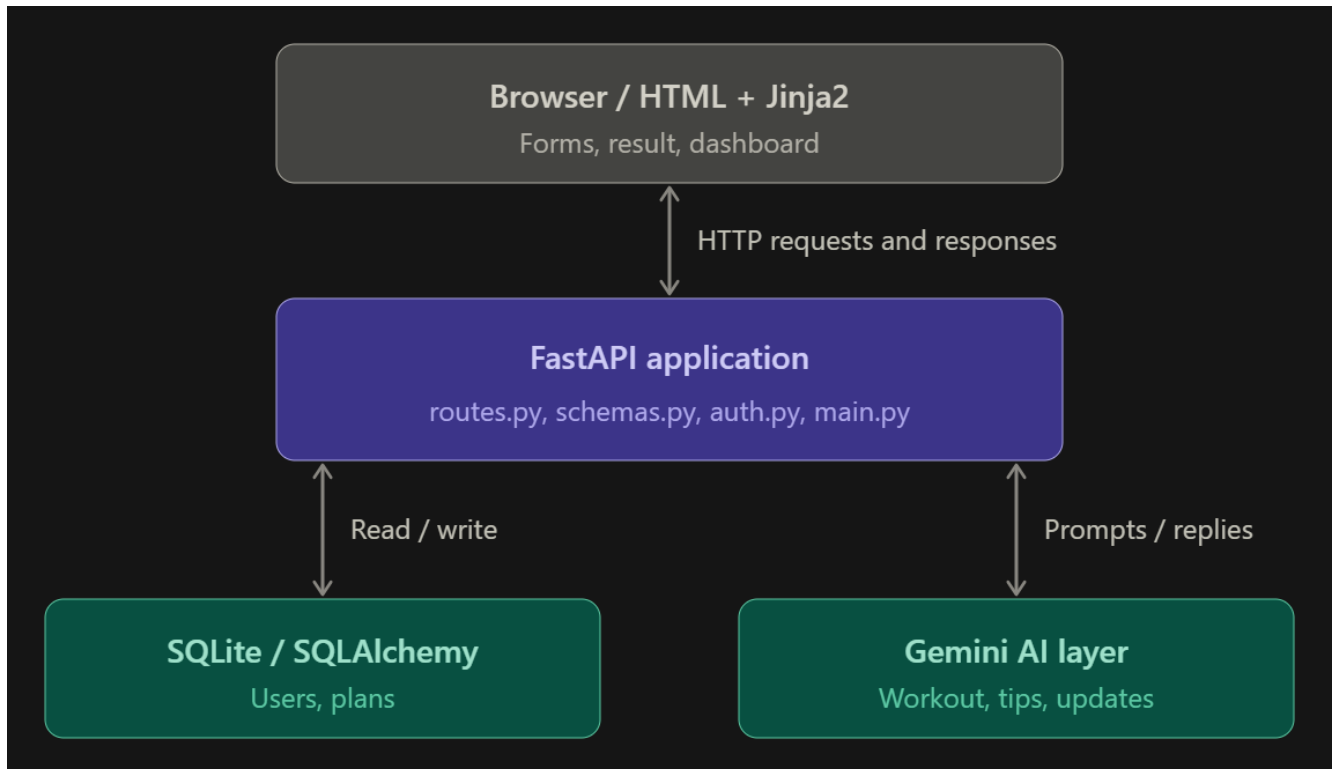


Solution Architecture

Date	03 October 2026
Team ID	FITBUDDY-TEAM
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	3 Marks

Architecture Overview



Module Responsibilities

Layer	Main Modules	Responsibility	Owner
Frontend	templates/, static/style.css, static/bmi.js	Forms, display, styling and client-side BMI support	Afraar M
API / Routing	main.py, routes.py	HTTP routes and application flow	Dinesh N
AI	gemini_generator.py, gemini_flash_generator.py, updated_plan.py, gemini_utils.py	Gemini prompts, generation and updates	Deepak B
Data	database.py, models.py, schemas.py	Persistence and data validation	Vinesh D + Dinesh N
Authentication	auth.py	User authentication and access control	Dinesh N

Data Flow

User input is validated → FastAPI invokes the required AI function → Gemini returns generated content → application stores the plan → Jinja2 renders the result → feedback can trigger a new AI generation.